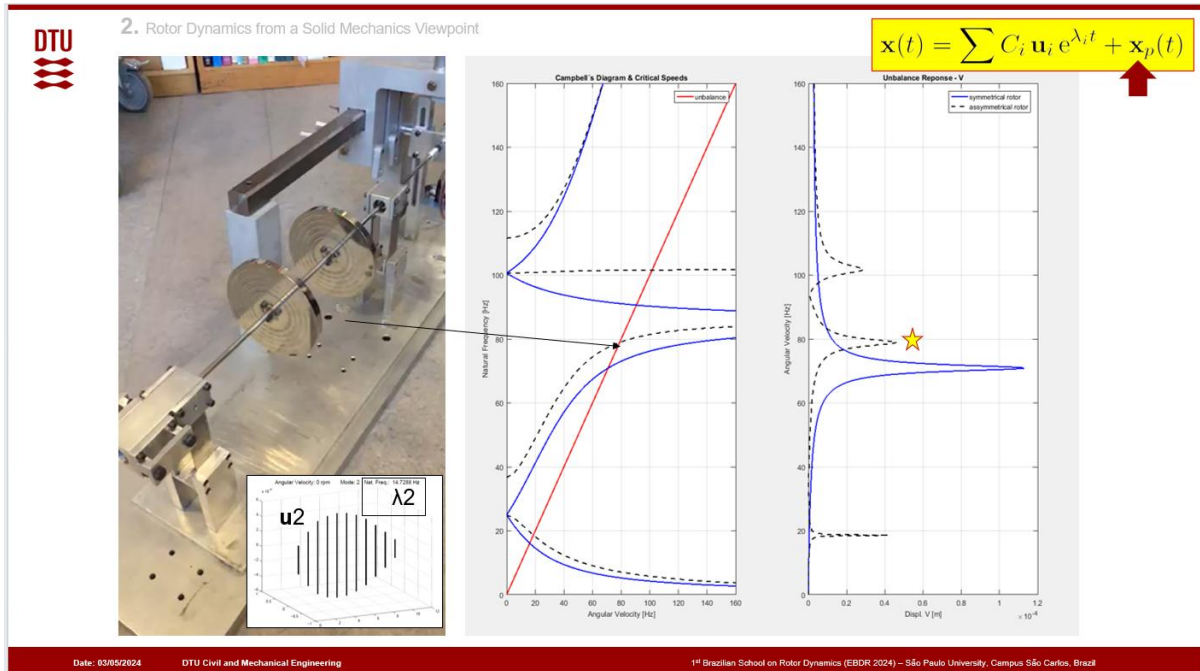


EXEMPLO 1 - Rotor em suportes elásticos assimétricos (undamped rotor system)

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5. Theoretical & Experimental Example (with Matlab)



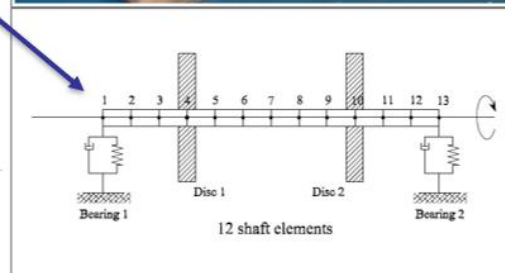
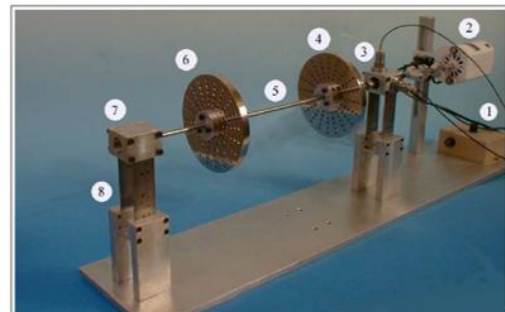
MATLAB Example : building the mechanical & finite element models

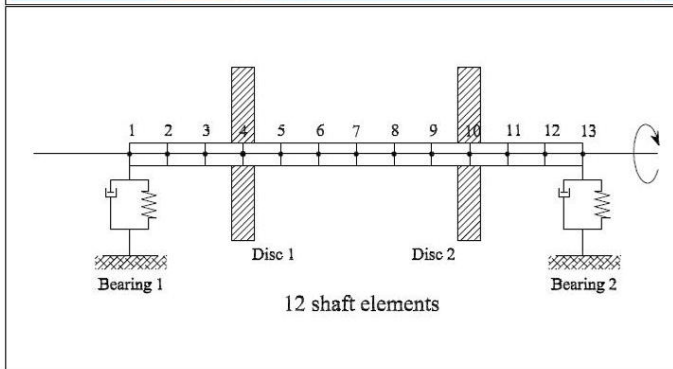
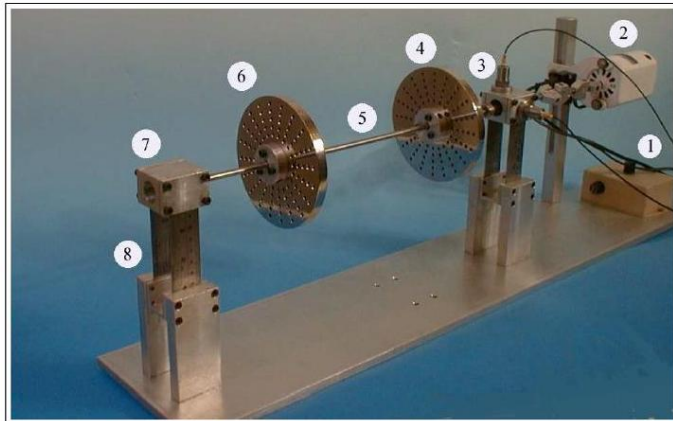
(flexible_rotor_model_DTU_c.m)

```

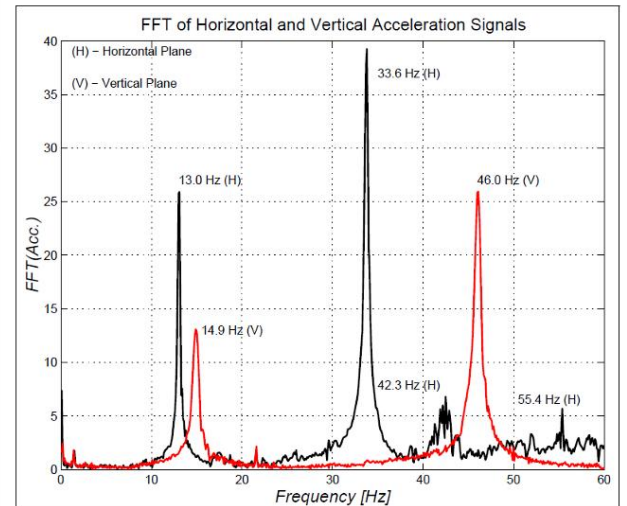
24 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
25 % DEFINITION OF THE STRUCTURE OF THE MODEL %
26 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
27 NE=12; % number of shaft elements
28 GL = (NE+1)*4; % number of degree of freedom
29 ND=2; % number of discs
30 NM=2; % number of bearings
31 CD1=4; % node - disc 1
32 CD2=10; % node - disc 2
33 CMM1=1; % node - bearing 1
34 CMM2=13; % node - bearing 2
35
36 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
37 % CONSTANTS %
38 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
39 E = 2.0E11; % {elasticity modulus [N/m^2]}
40 RAc0 = 7800; % {steel density [kg/m^3]}
41 RAl = 2770; % {aluminum density [kg/m^3]}
42
43 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
44 % OPERATIONAL CONDITIONS %
45 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
46 Omega = 0*2*pi; % angular velocity [rad/s]
47 Omegarm = Omega*60/2*pi;
48
49 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
50 % GEOMETRY OF THE ROTATING MACHINE %
51 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
52
53 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
54 % (A) DISCS %
55 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
56
57 Rd = 6/100; % disc radius [m]
58 espD = 1.1/100; % disc thickness [m]
59 MasD = pi*Rd^2*espD*RAc0; % disc mass [kg]
60 Id = 1/4*MasD*Rd^2+1/12*MasD*espD^2; % transversal mass moment of inertia of the disc [Kgm^2]
61 Ip = 1/2*MasD*Rd^2; % polar mass moment of inertia of the disc [Kgm^2]
62

```





MODE SHAPE	PLANE	Nat. Frequency (theoretical) [Hz]	Nat. Frequency (experimental) [Hz]	Error %
1	horizontal	13.8	13.0	6.1
2	vertical	14.7	14.9	1.3
3	horizontal	33.4	33.6	0.6
4	horizontal	42.3	42.3	0.0
5	vertical	47.1	46.0	2.1
6	horizontal	56.7	55.4	2.1
7	vertical	131.	(out of range)	—



Data :



flexible_rotor_modal_DTU_c



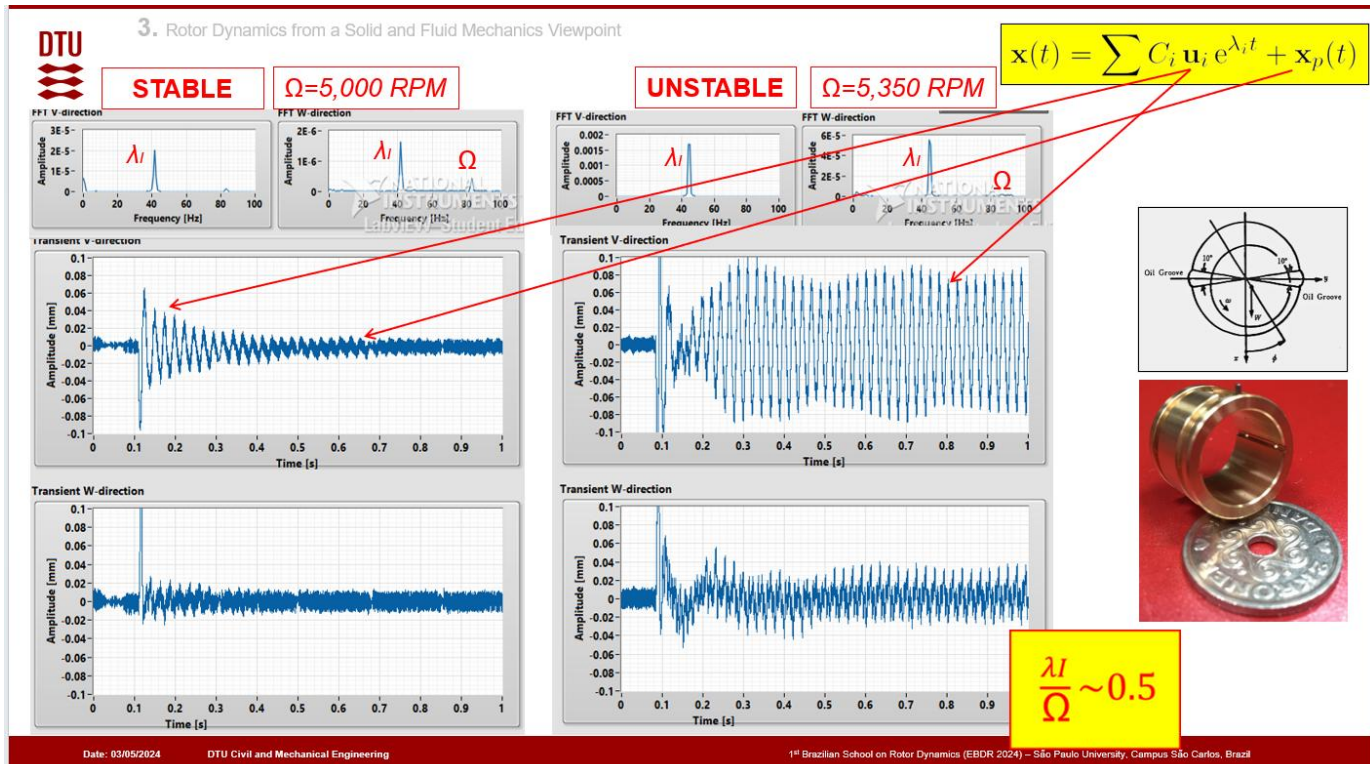
flexible_rotor_modal_DTU_c_24nodes



flexible_rotor_modal_DTU_c_48nodes

EXEMPLO 2 – Rotor em mancais hidrodinâmicos (damped and unstable system)

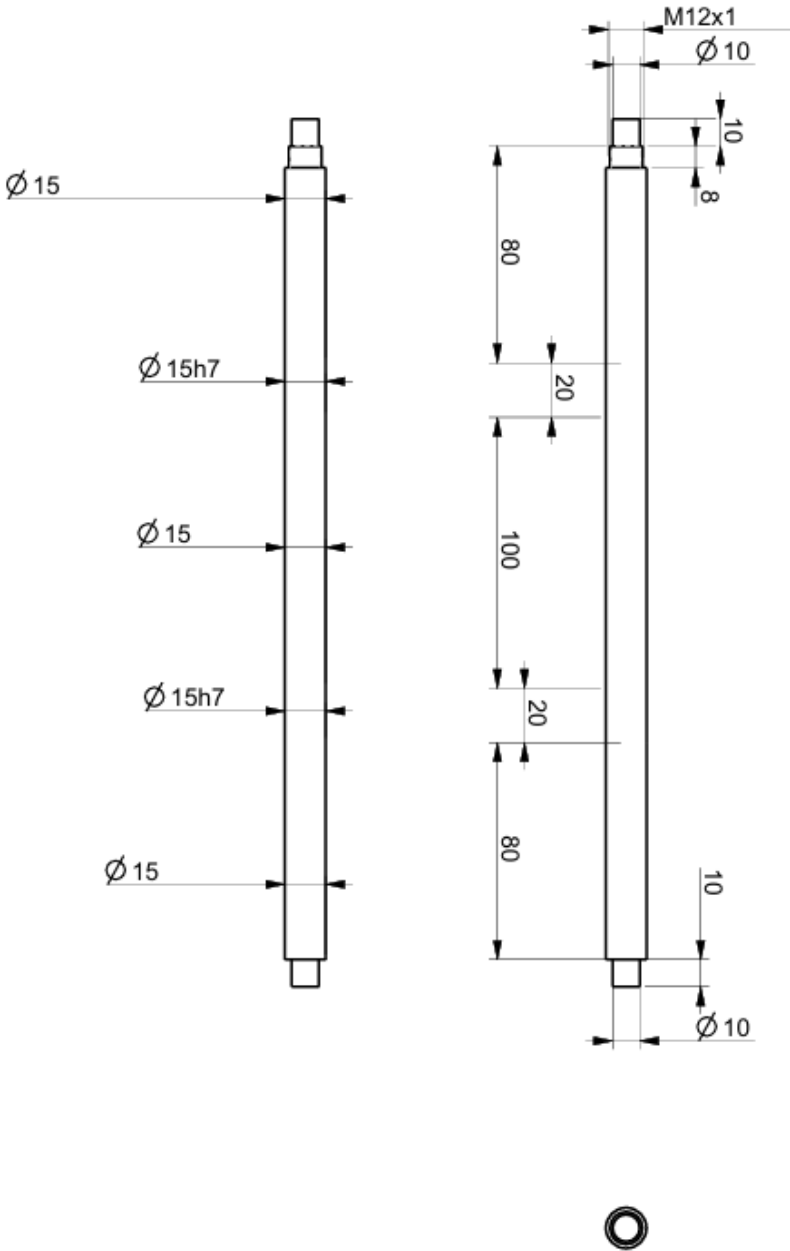
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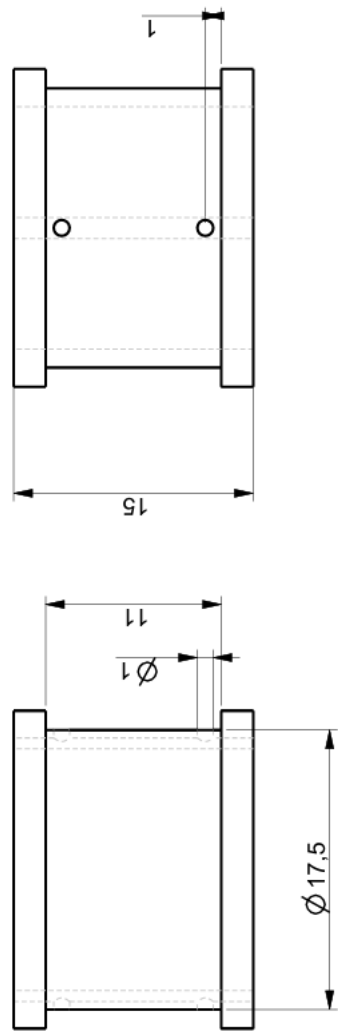
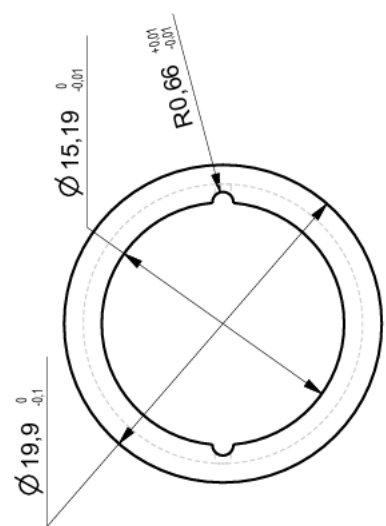
Data : mancal hidrodinâmico (tabela do Someya está na tese do bacharelado do Thomas e Lars.)

Eixo:

Axel	Bachelor	Stainless Steel	S103376	Tutor
Scale: 1:2	DTU MEK	1 stk	S103425	IFS

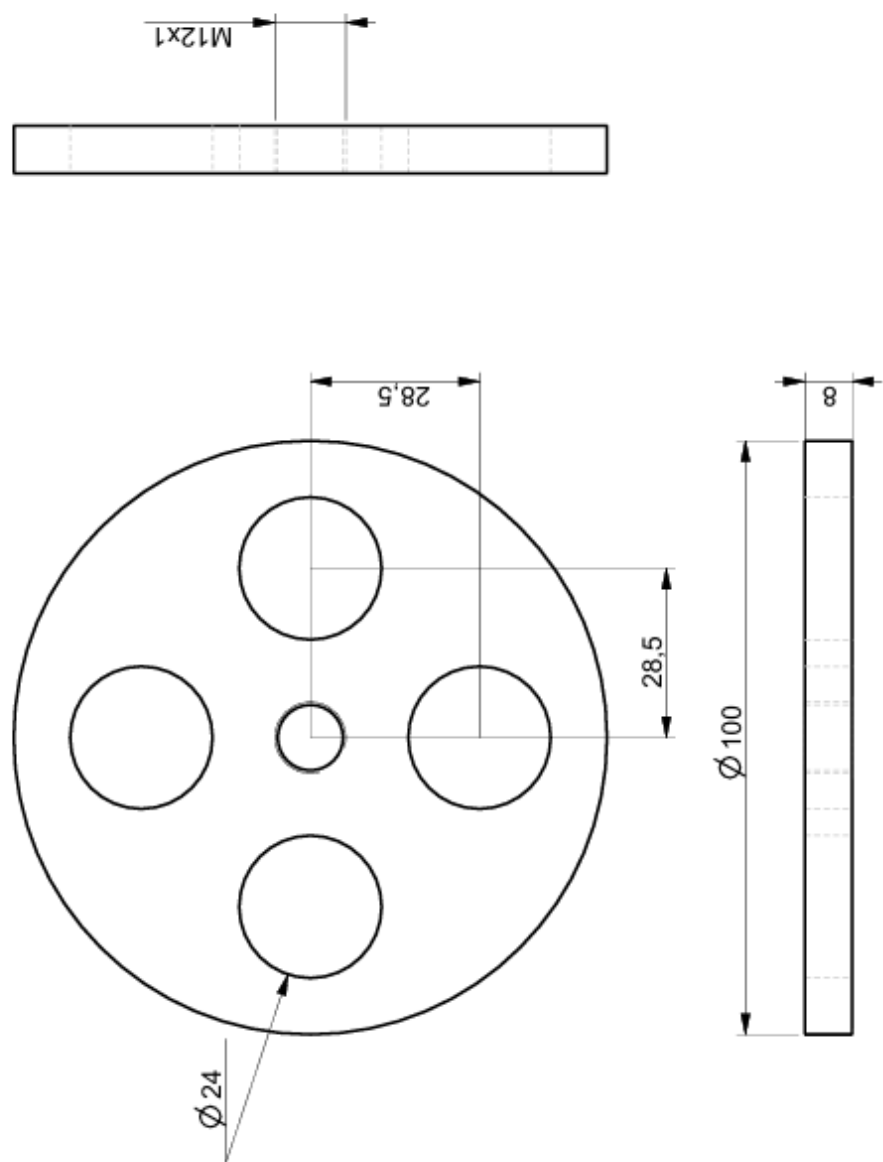


Mancal hidrodinâmico:



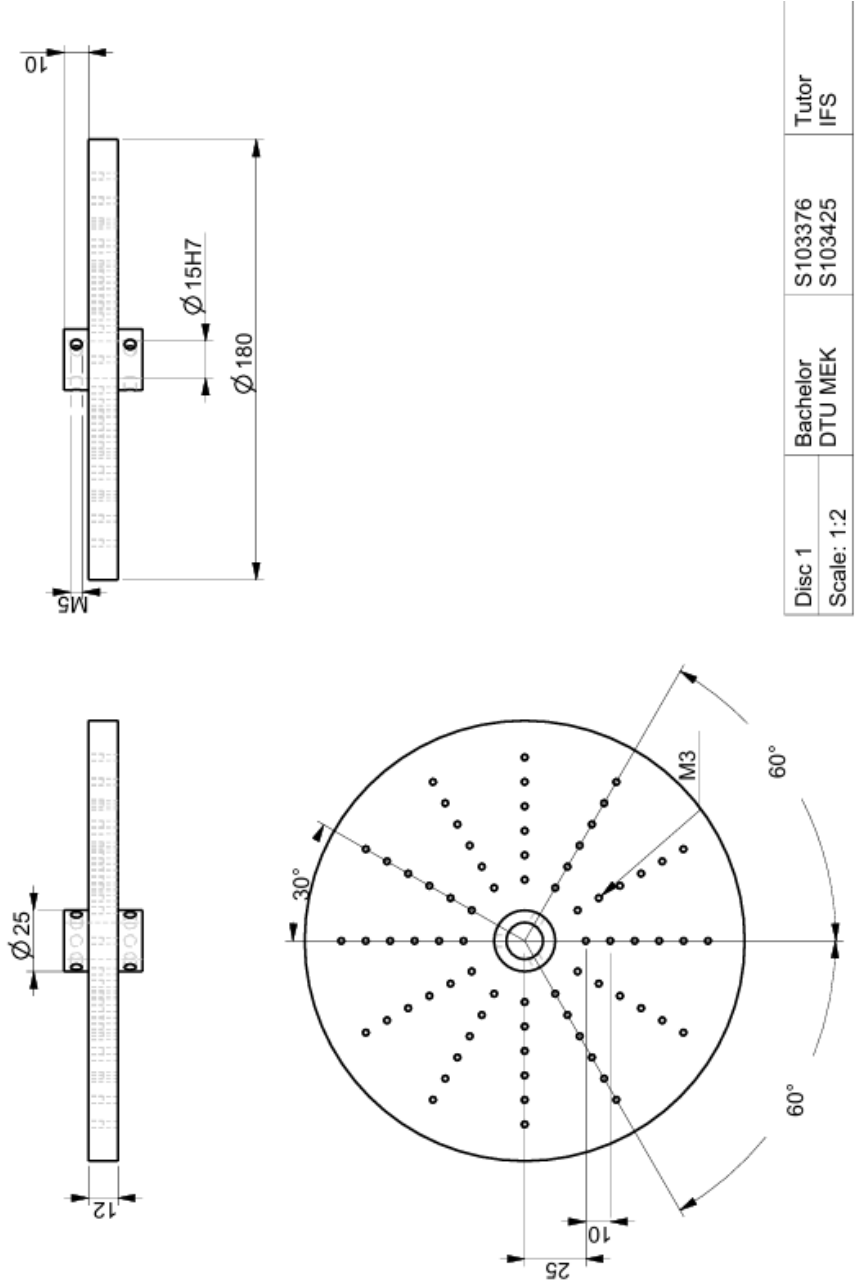
Journal Bearing	Bachelor DTU MEK	Brass	S103376 S103425	Tutor IFS
Scale: 1:3		4 stk		

OBS: Este disco foi feito de alumínio e sem os quatro furos! Disco utilizado para a medição das vibrações laterais do rotor !



Disc 2	Bachelor	Steel	Tutor
Scale: 1:1	DTU MEK	1 stk	IFS
			S103376
			S103425

OBS: Este é do disco de aço montado entre os dois mancais hidrodinâmicos.



Mais dados podem ser encontrados aqui. No apêndice, estão todos os desenhos da bancada.

Bachelor Thesis
Technical University of Denmark

Unstability in Hydrodynamical Bearings

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Lars Molsen, s103425



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